

COMP 3123 – Full Stack Development – Lab 1

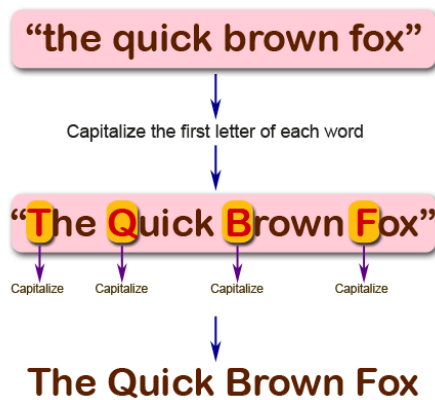
- JavaScript Refresher Exercises

Developer Note:

- Answer any 4 of the JavaScript exercises below
- Try to solve the problems without using search engines or stack overflow for the solutions.

Exercise 1:

Write a JavaScript program to capitalize the first letter of each word of a given string.



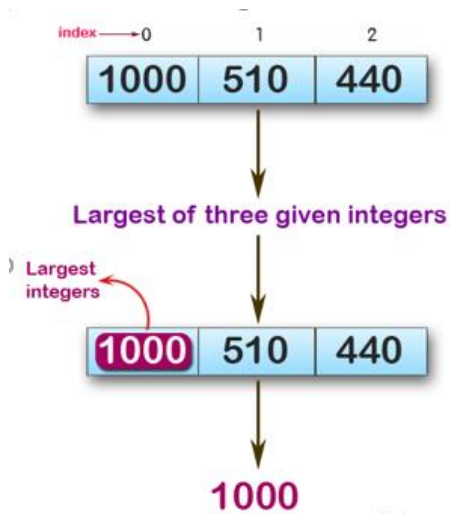
```
JS lab1.js x JS lab4-node-example.js
comp3123 > lab > labWeek1 > JS lab1.js > capitalizeSentence
1 // Exercise 1: Write a JavaScript program to capitalize the first letter of each word of a given string.
2
3 Windsurf: Refactor | Explain | Generate JSDoc | X
4 function capitalizeSentence(sentence) {
5   let words = sentence.split(" ");
6   let result = "";
7
8   for (let i = 0; i < words.length; i++) {
9     let word = words[i];
10    let capitalized = word.charAt(0).toUpperCase() + word.slice(1).toLowerCase();
11    result += capitalized + " ";
12  }
13
14  return result.trim();
15 }
16 console.log(capitalizeSentence("the quick brown fox"));
17
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL POLYGLOT NOTEBOOK PORTS Filter (e.g. text, !exclude, \

/Users/brenolopesmafra/.nvm/versions/node/v22.12.0/bin/node ./lab/labWeek1/lab1.js
The Quick Brown Fox

Exercise 2:

Write a JavaScript program to find the largest of three given integers.



```
console.log(max (1,0,1));  
console.log(max (0,-10,-20));  
console.log(max (1000,510,440));
```

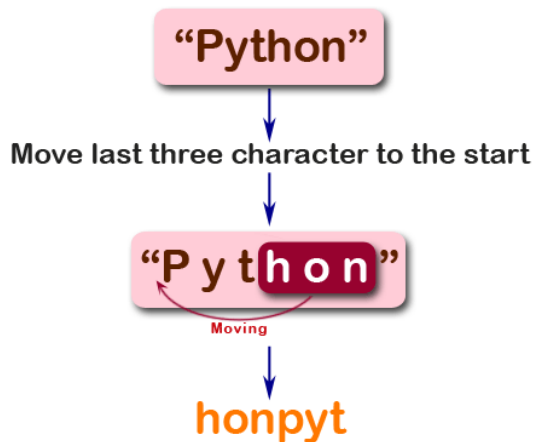
Sample Output:

```
1  
0  
1000
```

```
JS lab1.js x JS lab4-node-example.js  
comp3123 > lab > labWeek1 > JS lab1.js > findLargestNumber  
18 // Exercise 2: Write a JavaScript program to find the largest of three given integers.  
19  
20 function findLargestNumber(a, b, c) {  
21   if (a > b && a > c) {  
22     return a;  
23   } else if (b > a && b > c) {  
24     return b;  
25   } else {  
26     return c;  
27   }  
28 }  
29  
30 console.log(findLargestNumber(1000, 510, 440));  
31  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL POLYGLOT NOTEBOOK PORTS  
/Users/brenolopesmafra/.nvm/versions/node/v22.12.0/bin/node ./lab/labWeek1/lab1.js  
1000
```

Exercise 3:

Write a JavaScript program to move last three character to the start of a given string. The string length must be greater or equal to three.



```
console.log(right("Python"));  
console.log(right("JavaScript"));  
console.log(right("Hi"));
```

Sample Output:

honPyt

iptJavaScr

Hi

The screenshot shows a code editor with two tabs: "lab1.js" and "lab4-node-example.js". The active tab is "lab1.js", which contains the following code:

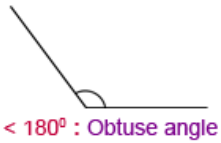
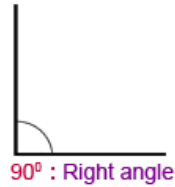
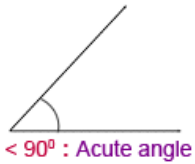
```
32 // Exercise 3: Write a JavaScript program to move last three character to the start of a given string. The  
33  
34 function moveLastThreeCharactersToStart(str) {  
35   if (str.length >= 3) {  
36     return str.slice(-3) + str.slice(0, -3);  
37   } else {  
38     return str;  
39   }  
40 }  
41  
42 console.log(moveLastThreeCharactersToStart("Python"));
```

The output of the code is shown in the terminal at the bottom:

```
./Users/brenolopesmafra/.nvm/versions/node/v22.12.0/bin/node ./Lab/labWeek1/lab1.js  
honPyt
```

Exercise 4:

Write a JavaScript program to find the types of a given angle.



```
console.log(angle_Type(47))  
console.log(angle_Type(90))  
console.log(angle_Type(145))
```

Sample Output:

Acute angle

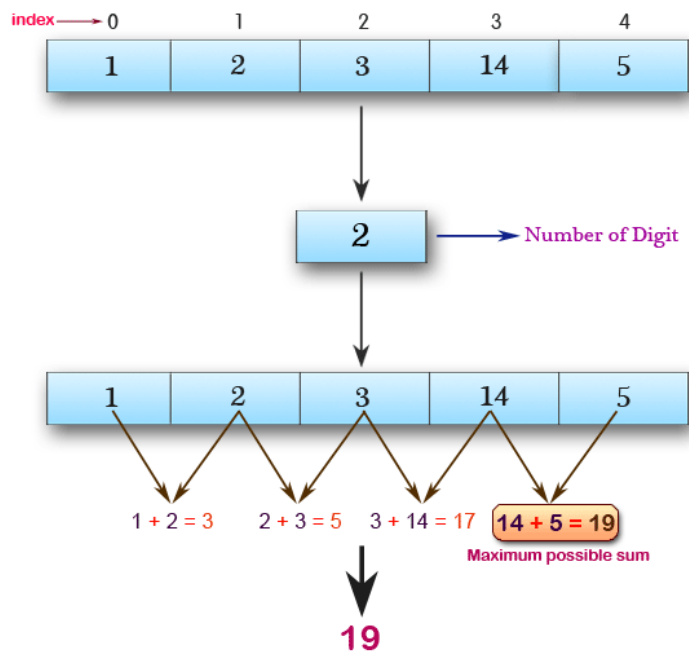
Right angle

Obtuse angle

```
JS lab1.js x JS lab4-node-example.js  
comp3123 > lab > labWeek1 > JS lab1.js > findAngleType  
43  
44 // Exercise 4: Write a JavaScript program to find the types of a given angle.  
45 // • Acute angle: An angle between 0 and 90 degrees.  
46 // • Right angle: An 90 degree angle.  
47 // • Obtuse angle: An angle between 90 and 180 degrees.  
48 // • Straight angle: A 180 degree angle.  
49  
Windsurf: Refactor | Explain | Generate JSDoc | X  
50 function findAngleType(angle) {  
51   if (angle > 0 && angle < 90) {  
52     return "Acute angle";  
53   } else if (angle === 90) {  
54     return "Right angle";  
55   } else if (angle > 90 && angle < 180) {  
56     return "Obtuse angle";  
57   } else if (angle === 180) {  
58     return "Straight angle";  
59   } else {  
60     return "Invalid angle";  
61   }  
62 }  
63  
64 console.log(findAngleType(47));  
65  
66  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL POLYGLOT NOTEBOOK PORTS  
Filter (e.g. text, lexclude, \n)  
/Users/brenolopesmafra/.nvm/versions/node/v22.12.0/bin/node ./lab/labWeek1/lab1.js  
Acute angle
```

Exercise 5:

Write a JavaScript program to find the maximum possible sum of some of its k consecutive numbers (numbers that follow each other in order.) of a given array of positive integers.



```
console.log(array_max_sum([1, 2, 3, 14, 5], 2))
console.log(array_max_sum([2, 3, 5, 1, 6], 3))
console.log(array_max_sum([9, 3, 5, 1, 7], 2))
```

Sample Output:

19
12
12